

Math 526, S05
Quiz 1

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$ and $\mathbf{C} = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}$.

a) Compute $\mathbf{AB}$ and $(\mathbf{AB})^t$.

Solution:

$$\mathbf{AB} = \begin{bmatrix} -4 & -2 & 0 \\ -7 & -5 & -3 \\ -10 & -8 & -6 \end{bmatrix}$$

$$(\mathbf{AB})^t = \begin{bmatrix} -4 & -7 & -10 \\ -2 & -5 & -8 \\ 0 & -3 & -6 \end{bmatrix}$$

b) Compute $\mathbf{A}^t\mathbf{B}^t$. Does $(\mathbf{AB})^t = \mathbf{A}^t\mathbf{B}^t$?

Solution:

$$\mathbf{A}^t = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}, \quad \mathbf{B}^t = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

$$\mathbf{A}^t\mathbf{B}^t = \begin{bmatrix} -10 & -4 & 2 \\ -11 & -5 & 1 \\ -12 & -6 & 0 \end{bmatrix}$$

So $(\mathbf{AB})^t \neq \mathbf{A}^t\mathbf{B}^t$.

c) Compute $\mathbf{C}^2$ and $\mathbf{C}^3$.

Solution:

$$\mathbf{C}^2 = \begin{bmatrix} 4 & -9 \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{C}^3 = \begin{bmatrix} 8 & -21 \\ 0 & 1 \end{bmatrix}$$

Problem 2. (10 points) Let $\mathbf{A} = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$.

a) Find the $\mathbf{L} - \mathbf{U}$ factorization of $\mathbf{A}$.

Solution:

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ -2 & & \\ 3 & & \end{bmatrix} \quad \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & -6 & 5 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ -2 & & \\ 3 & 3 & \end{bmatrix} \quad \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}$$

So we have

$$\mathbf{L} = \begin{bmatrix} 1 & & \\ -2 & 1 & \\ 3 & 3 & 1 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}.$$

b) Using the above factorization to solve $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

Solution:

First step: Solve $\mathbf{Ly} = \mathbf{b}$.

$$\begin{bmatrix} 1 & & \\ -2 & 1 & \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$$

We get

$$y_1 = 2, \quad y_2 = 6, \quad y_3 = -20.$$

Second step: Solve $\mathbf{Ux} = \mathbf{y}$.

$$\begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ -20 \end{bmatrix}$$

We have

$$x_3 = -1, \quad x_2 = -\frac{1}{2}, \quad x_1 = \frac{1}{2}.$$

So we have

$$\mathbf{x} = \begin{bmatrix} 1/2 \\ -1/2 \\ -1 \end{bmatrix}.$$